4. The mean height of 11 students in a group is 150 cm. The heights of the students are 154 *cm*, 145 *cm*, Y *cm*, Y + 4 *cm*, 160 *cm*, 151 *cm*, 149 *cm*, 149 *cm*, 150 *cm*, 144 *cm* and 140 *cm*. Find the value of Y and the heights of two students?

5. The mean of runs scored by a cricket team in the last 10 innings is 276. If the scores are 235, 400, 351, x, 100, 315, 410, 165, 260, 284, then find the runs scored in the fourth innings.

6. Find the mean of the following data.

   5.1, 4.8, 4.3, 4.5, 5.1, 4.7, 4.5, 5.2, 5.4, 5.8, 4.3, 5.6, 5.2, 5.5

7. Arithmetic mean of 10 observations was found to be 22. If one more observation 44 was to be added to the data, what would be the new mean?

Objective type questions

8. _____ is a representative value of the entire data.

   (i) Mean (ii) range (iii) minimum value (iv) maximum value

9. The mean of first fifteen even numbers is _____

   (i) 4 (ii) 16 (iii) 5 (iv) 10

10. The average of two numbers are 20. One number is 24, another number is _____

    (i) 16 (ii) 26 (iii) 20 (iv) 40

11. The mean of the data 12, x, 28 is 18. Find the value of x.

    (i) 18 (ii) 16 (iii) 14 (iv) 22

5.6 Mode

As we have discussed earlier that the arithmetic mean is one of the form of representative value or measures of central tendency of a group of data. Depending upon the data and its purpose, other measures of central tendency may be used.

Consider the example of sale details of different sizes of footwear in a shop for a week.

Size of the footwear	5"	6"	7"	8"	9"	10"	11"
Number of footwear sets sold	20	35	16	65	32	25	10

The shopkeeper has to replenish his stock at the end of the week. Suppose we find the arithmetic mean of the footwear sold,

$$\text{Mean} = \frac{20+35+16+65+32+25+10}{7} = \frac{203}{7} = 29.$$

Average number of footwear is 29. This means that the shopkeeper has to get 29 pairs of footwear in each size. Will it be wise to decide like this?

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